

Fire Fito

Arduino-Based Fire Fighting Robot Project Report

Abstract

Fire Fito is an Arduino UNO based autonomous fire fighting robot designed to detect and extinguish fire automatically. The robot uses flame sensors to identify the presence and direction of fire. Once detected, the robot moves toward the fire source and activates a water pump or fan to extinguish the flame. The project demonstrates the application of embedded systems, robotics, automation, and sensor integration in safety systems.

Introduction

Automation and robotics play an important role in modern safety systems. Fire accidents can cause severe damage to life and property. Fire Fito is designed as a smart robotic solution capable of reducing human risk during small-scale fire incidents. This project uses Arduino UNO as the central controller and integrates motors, sensors, and extinguishing systems to create an autonomous robot.

Objectives

1. Detect fire using flame sensors.
2. Move autonomously toward the fire source.
3. Activate water pump automatically.
4. Demonstrate embedded system control using Arduino UNO.
5. Create a low-cost and effective safety robot.

Components Required

Arduino UNO, L298N Motor Driver, Flame Sensors, DC Motors, Water Pump, Servo Motor (optional), Chassis, Battery Pack, Jumper Wires, Wheels, Breadboard, Relay Module.

Working Principle

The flame sensors continuously monitor infrared radiation emitted by flames. When the sensors detect fire, the Arduino UNO processes the signal and determines the direction of the flame. Based on the sensor input, the robot changes its movement direction using DC

motors controlled by the L298N motor driver. Once the robot reaches the fire source, it stops and activates the water pump to extinguish the fire.

Circuit Diagram Explanation

The flame sensors are connected to analog pins A0, A1, and A2 of the Arduino UNO. The motor driver is connected to digital output pins for controlling motor direction. The water pump is connected through a relay or transistor circuit. Power is supplied using a battery pack. The Arduino acts as the brain of the system.

Advantages

1. Reduces human involvement in dangerous situations.
2. Low-cost and easy to build.
3. Useful for learning robotics and automation.
4. Autonomous operation.
5. Portable and compact design.

Limitations

1. Limited water storage capacity.
2. Effective only for small fires.
3. Sensor accuracy may be affected by environmental light.
4. Battery backup limitations.
5. Cannot handle industrial-scale fires.

Applications

1. Home safety systems.
2. Laboratories.
3. Small industries.
4. Educational robotics projects.
5. Research and automation demonstrations.

Future Improvements

Future versions of Fire Fito can include IoT monitoring, Bluetooth or Wi-Fi control, AI-based fire detection using cameras, obstacle avoidance sensors, GPS tracking, and real-time alerts using mobile applications.

Conclusion

Fire Fito demonstrates the practical implementation of robotics and embedded systems for safety applications. The project is affordable, educational, and highly useful for understanding automation concepts. It can serve as a foundation for advanced fire safety robots in the future.

Block Diagram

